Welcome to

Essentials of Biomedical Informatics

Foundations, OMOP, SQL, and Reproducible Analysis

A Self-Paced, Asynchronous Module

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Welcome

Welcome to Essentials of Biomedical Informatics! Over the next four weeks, you'll go from installing your first tools to building and analyzing a real patient cohort using the same data standard (OMOP) that powers national research networks like the All of Us Research Hub, the National Clinical Cohort Collaborative (N3C), and The National Patient-Centered Clinical Research Network (PCORnet).

No prior programming experience is assumed. Every concept is introduced step by step, and every exercise includes a fully worked example before asking you to try it yourself. Every tool you'll use is free and open-source, so whatever you learn here, you can keep using long after this course ends.

How This Course Works

- This module is self-paced and asynchronous — there are no live sessions to attend.
- Plan for roughly 3–4 hours per week.
- Each week has its own Module Guide (a PDF document) covering that week's concepts, software setup, and exercise instructions.
- A self-check quiz before submitting your work, and the grading rubric are available on each week's module for reference
- Each week also has one or more companion code files (R and/or Python, and SQL from Week 2 onward). Every code file follows the same two-part structure:

Part	What it is
Part 1 — Walkthrough	A fully worked example. Read it, run every step, and make sure you understand what each piece does before moving on.
Part 2 — Your Turn	A short exercise you complete yourself. This is the graded deliverable for the week, submitted via the corresponding Brightspace dropbox.

Each week is written to stand on its own — if life gets in the way and you fall behind, you can always pick up the current week's guide and get oriented quickly, rather than needing to catch up from the very beginning.

What You'll Build, Week by Week

Week	Focus	What you'll do
1	Foundations	Install R, Python, and Quarto; explore a small synthetic patient dataset; produce your first reproducible report.
2	The OMOP Common Data Model	Learn how health data gets standardized across institutions; write your first SQL queries against a realistic (synthetic) clinical database.
3	Cohort Building with SQL	Use CTEs and window functions to define a real patient cohort by diagnosis and treatment history; connect that SQL to R and Python.
4	Wrangling & Visualization	Join in clinical measurements, handle missing data honestly, build a "Table 1," and visualize a trend over time.

By the end of this course, you'll have taken a single research question all the way from raw, standardized data to a finished, reproducible analysis — in both R and Python.

Tools You'll Need (All Free)

- R and RStudio Desktop
- Python (via the free Miniconda distribution)
- Quarto, for rendering reproducible reports
- DuckDB, a lightweight SQL database engine (introduced in Week 2 — nothing to install before then)

Detailed, step-by-step installation instructions are in the Week 1 Module Guide — you don't need to install anything before then.

A Note on the Data

Every patient dataset used in this block is entirely synthetic (randomly generated by a computer, not drawn from real people). This lets you practice every technique safely and without any privacy concerns, before you go on to work with real, IRB-approved data later in your training or research.

How You'll Be Assessed

Each week has one graded deliverable: the “Your Turn” section of that week's exercise, submitted via the matching Brightspace dropbox. Self-check quizzes in each Module Guide are for your own review and are not graded. See each week's section for the specific rubric.

Getting Help

Because this course is self-paced, reaching out as soon as you're stuck, rather than waiting, makes the biggest difference. Feel free to reach out to my email (cfiguero@mcw.edu) or office phone number (414-955-0500). I expect the turnaround time to be within 24 hours.

Ready to Start?

Open the Week 1 Module Guide and begin. Welcome aboard — we're glad you're here.